

HW13

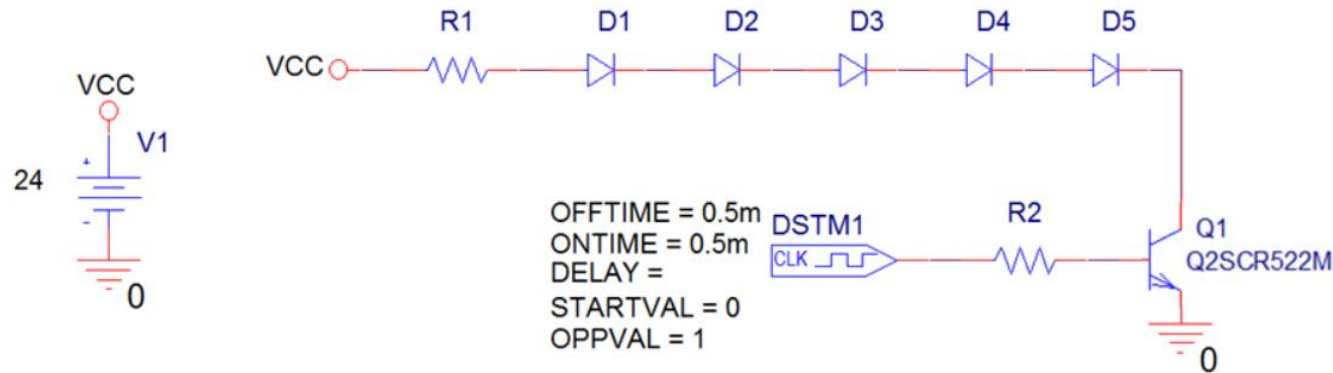
엄민호

1. Ambient LED 회로 설계

LED : SMLE13WBC8W1 (Rohm)

Control device : npn(Q2SCR522M, Rohm)

VCC : 24V, I at LED : 5mA, digiclock(1kHz, duty = 50%)

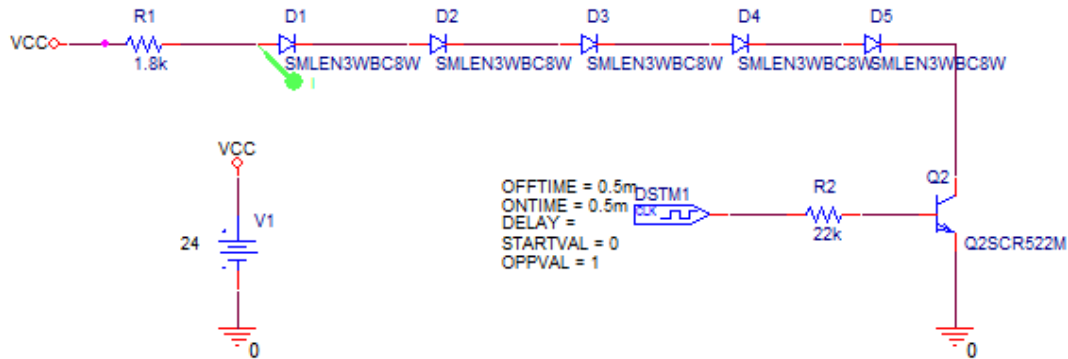


LED SMLE13WBC8W1 (Rohm) 소자 단종

➔ Rohm 사의 대체품 SMLN3WBC8W 채택 spec 거의 똑같음

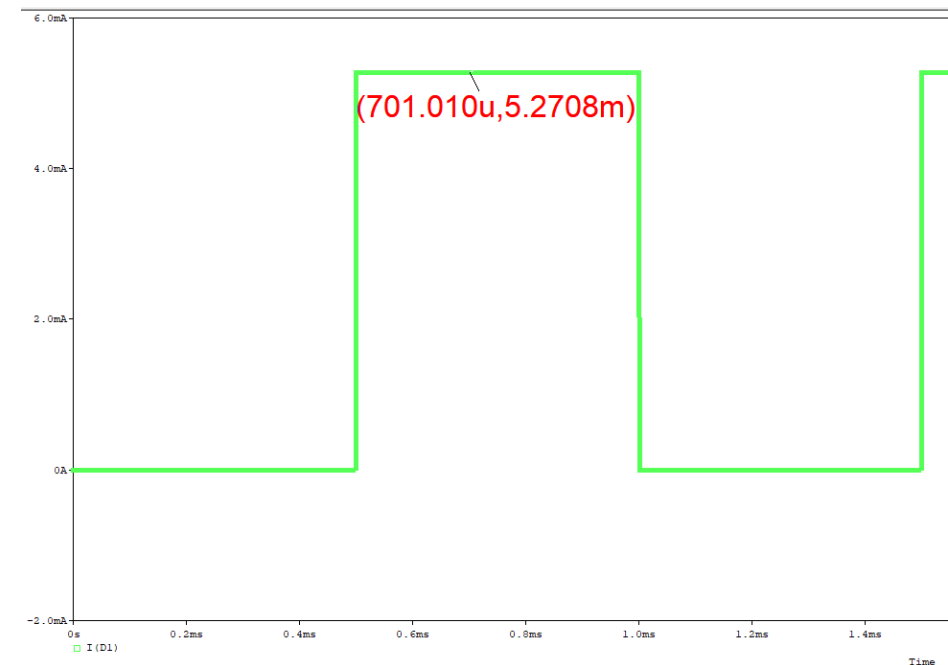
Part Number	 SMLE13WBC8W	 SMLN3WBC8W
Ordering Part Number	SMLE13WBC8W1	SMLN3WBC8W1
Similar Level	-	Same Pinout, Package
Data Sheet		
Supply Status	Not Recommended for New Designs	Recommended
Package	SML-E1	SMLN
Unit Quantity	5000	5000
Minimum Packing Quantity	5000	5000
Packing Type	Taping	Taping
RoHS	Yes	Yes
Package Type	Mini-mold	Mold
LED Type	EXCELED	Ultra-thin
Application	-	Smart Phone & Mobile
Package size[mm]	1.6x0.8 (t=0.36)	1.6x0.8 (t=0.36)
Emitting color	White	White
Luminous Intensity(Single)(Typ.)[mcd]	120.0	120.0
Number of Brightness Rank	-	3
Forward Voltage V F (Typ.) [V]	2.9	2.9
I F @ V F [mA]	-	5
Chromaticity coordinates(x,y)	(0.30, 0.30)	(0.30, 0.30)
Chip Structure	InGaN	InGaN
Power Dissipation [mW]	33	33
Operating Temperature[°C]	-40 to 85	-40 to 85
Storage Temperature [°C]	-	-40 to 100

1. Ambient LED 회로 설계

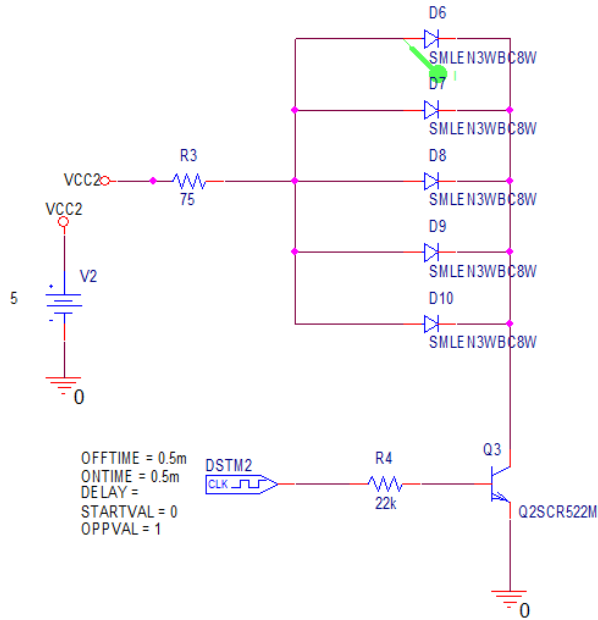


- R2 값 수십 K옴 -> 22kΩ 선택
- SMLE13WBC8W Datasheet의 $V_f = 2.9V$
-> 5개 직렬 -> 14.5V 전압 강하
- Q2SCR522M Datasheet의 $V_{ce.sat} = 0.2V$ 가정
- $R1 = \frac{V_{cc} - (2.9 \times 5) - 0.2}{I_d} = \frac{24 - 14.5 - 0.2}{5m} = 1.86k \Omega$
➔ Standard resistor 1.8k Ω 채택

I at LED : 5.27mA -> Simulation 만족



2. Ambient LED 회로 설계 (When VCC : 5V)



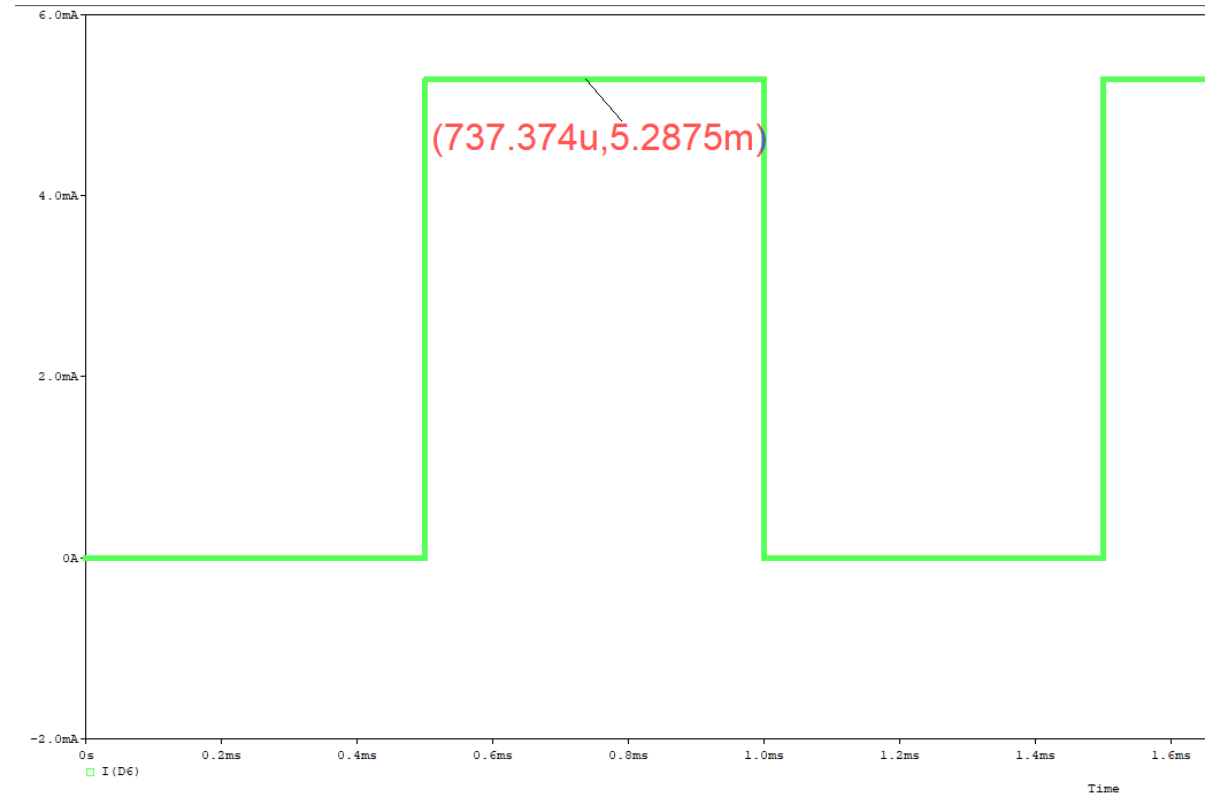
VCC : 5V -> 이므로 LED5개를 병렬로 연결

I_{R3} : 25mA -> each LED : 5mA

$$R3 = \frac{VCC - V_f - V_{CE.SAT}}{25m} = \frac{5 - 2.9 - 0.1}{25m} = 80\Omega$$

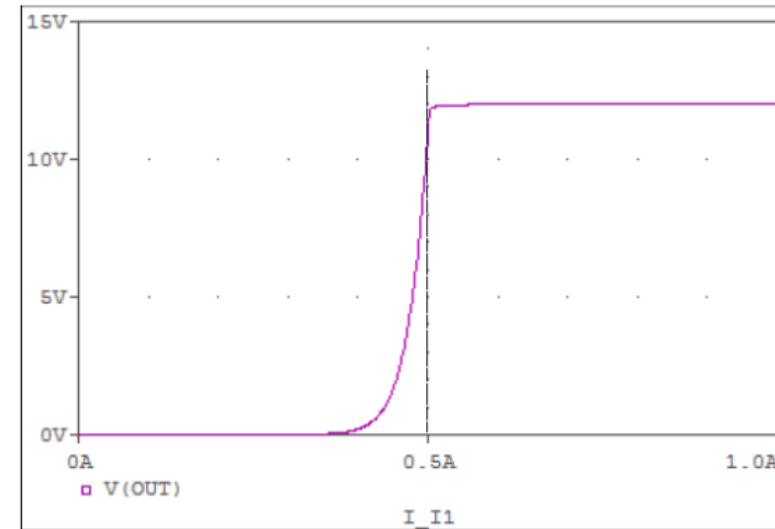
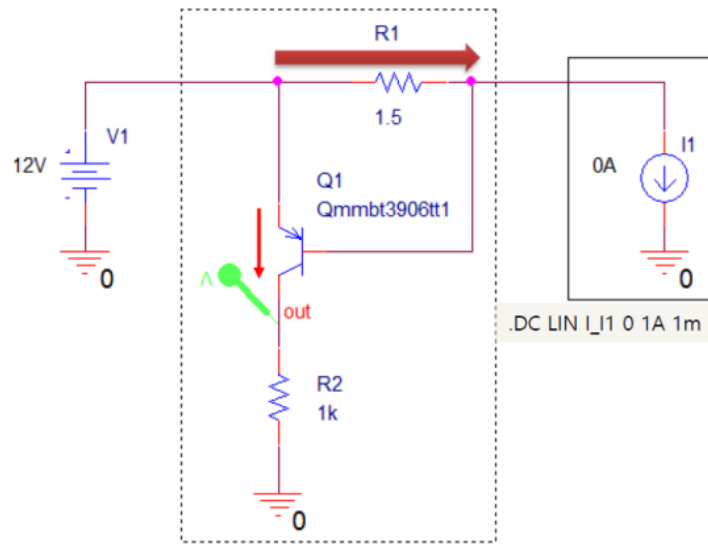
➔ Standard resistor : 75 Ω 선정

I at LED : 5.29mA -> Simulation 만족



over current detector output voltage level change

(range : 0 ~12v -> 0~5V)



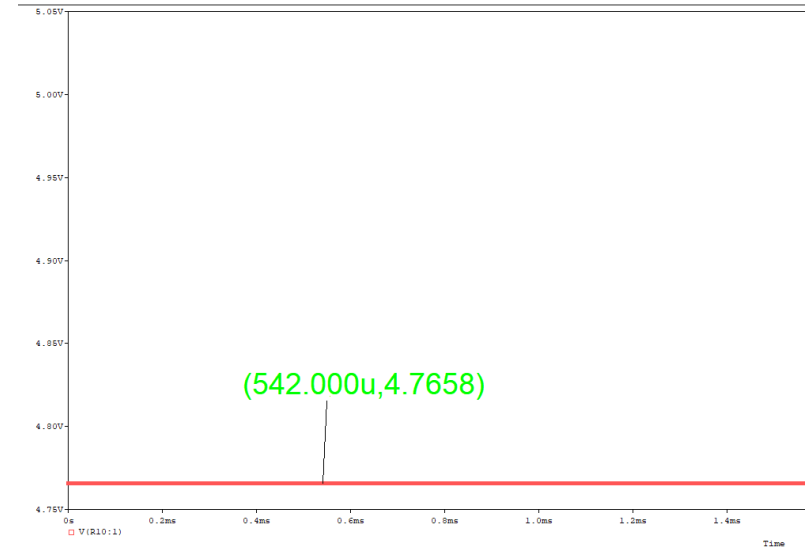
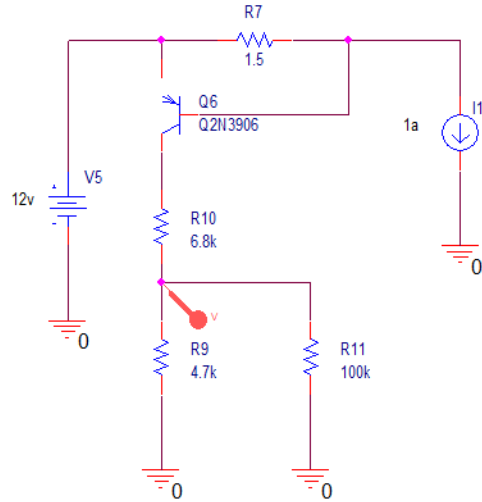
Qmmbt3906tt1 : $V_{BE.on} = 0.75V \rightarrow R_s = 1.5 \rightarrow 0.5A$ over current detector

MCU의 input dynamic range = 0~5V

MCU의 입력저항(Input impedance) 은 보통 수백 k Ω ~ 수 M Ω
->저항을 이용한 전압 분배 선택하여 output voltage level change

over current detector output voltage level change

(range : 0 ~12v -> 0~5V)



MCU input resistance : 100k Ω 가정
Standard resistance : 6.8k Ω , 4.7k Ω 선택

Simulation -> output voltage : 4.76V 만족